

$$\{1, \dots, n\}$$

Assegnato

$$\{i_1, \dots, i_k\} \in C_{n,k}$$

$\wedge$
 $\{1, \dots, n\}$

Permutando gli elementi di $\{i_1, \dots, i_k\}$
trovo $k!$ elementi di $D_{n,k}$.

Inoltre ogni elemento di $D_{n,k}$ può essere
visto come una permutazione di un qualche
elemento di $C_{n,k}$.

$$\# C_{n,k} \cdot k! = \# D_{n,k}$$

$$(i_1, \dots, i_k) \in D_{n,k} \rightsquigarrow \underbrace{\{j_1, \dots, j_k\}}_{\text{ordinamento}}$$

$$n=4$$

$$k=3$$

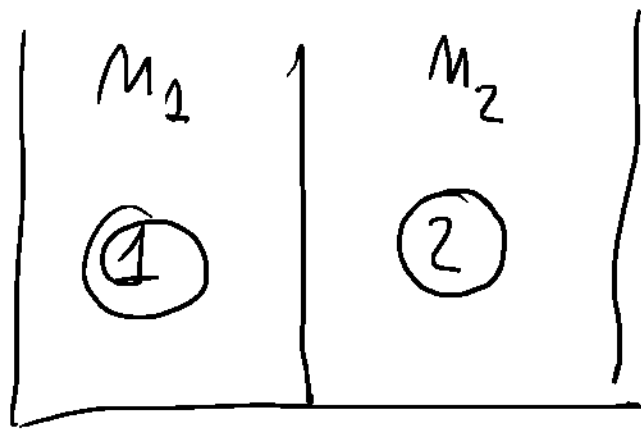
$$\{1, 2, 3, 4\}$$

$$D_{4,3} = \{ \underbrace{(1,2,3)}_{\substack{\{ \\ (1,3,2) \\ (2,1,3) \\ (2,3,1) \\ (3,2,1) \\ (3,1,2) \}}, \underbrace{(1,2,4)}_{\{}, \underbrace{(1,3,4)}_{\{}, \underbrace{(2,3,4)}_{\{}$$

$$\#D_{4,3} = 24$$

$$\text{In effect: } \#D_{4,3} = \frac{4!}{(4-3)!} = \frac{4!}{1!} = 4 \cdot 3 \cdot 2 = 24.$$

$$C_{4,3} = \{ \{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\} \}. \quad \#C_{4,3} = \frac{4!}{3!(4-3)!} = 4$$



$$M_1 = 3$$

$$M_2 = 2$$

$$M_1 + M_2 = 5$$

$$M = 3$$

$$\{ \underbrace{1, 2, 3}_{\textcircled{1}}, \underbrace{4, 5}_{\textcircled{2}} \}$$

P_k con $k=1$?

$P(\text{1 oggetto } \textcircled{1} \text{ e } 2 \text{ oggetti } \textcircled{2}) = ?$

$$\{1, 4, 5\}, \{2, 4, 5\}, \{3, 4, 5\}$$

$$\frac{\binom{3}{1} \binom{2}{2}^{\leq 1}}{\binom{5}{3}} = \frac{3}{10}$$